Tsng, Kahiok

Taipei, Taiwan | tsng@kahiok.com | github.com/madmaxieee | www.linkedin.com/in/kahiok | www.kahiok.com

Education

National Taiwan University Taipei, Taiwan
Bachelor of Science in Engineering, Electrical Engineering *Sep 2020 — June 2024*
- GPA: 3.86/4.3

Work Experience

Firmware / System Software Engineer Aug 2024 — Present
Google *New Taipei, Taiwan*
- Architected a modular Modem BSP test platform (36 classes, 137 checks) with DAG filtering and crash telemetry, running 800+ sessions (1,330+ device-hrs) to offload ~1,060 engineer-hrs, cut triage by 85% (20m to 3m), and catch 17+ defects.
- Isolated shared-PHY PLL race conditions causing kernel panics during concurrent Wi-Fi/Modem PCIe ASPM (L1/L1.2), building a 400+ case concurrency suite (1,500+ device-hrs) that cut root-cause repro by >98% (>100h to <2h).
- Engineered an AT command serial relay (`at_relay`) in 3 days to unblock AT&T and NTN Skylo lab certifications, scaling it into a production Linux userspace daemon with async URC routing across Telephony, QA, and Factory workflows.
- Modernized Android modem userspace across 80+ CLs—migrating 10+ vendor daemons to Soong and Rust (`rdroidtest`), upgrading logging to AIDL V3 schemas, and hardening NVRAM reset with a scoped Rust service.
- Consolidated 8+ out-of-tree WWAN kernel driver repos into a single-tree modular Kbuild architecture with automated presubmits, and spearheaded strict SELinux enforcing mode across 10+ modem daemons with zero disruptions.
- Standardized Nix/Clangd kernel and Android dev environments (saving 2–3 onboarding days/engineer), built an org-wide 3-way merge tool (`unclash.nvim`), and mapped vendor daemon/PCIe state machines for 3-day userspace bring-up.

Software Engineering Intern, Frontend Nov 2023 — Mar 2024
Appier *Taipei, Taiwan*
- Engineered the LINE Beacon configuration UI and workflow in the marketing console for proximity-targeted campaigns
- Collaborated with backend engineers to integrate RESTful APIs, client state management, and form validation for launch
- Partnered with product managers to diagnose and resolve client-facing production UI defects across enterprise accounts

Projects

Creator & Contributor, Open-Source CLI & Neovim Tooling (github.com/madmaxieee) 2024 — Present
- Built `loglit` (49 stars, Go), a CLI log highlighter with regex matching and dual-TTY stderr peeking for piped streams
- Authored `fff-snacks.nvim` (39 stars, 8 forks) and `unclash.nvim` (22 stars, 3-way merge conflict manager) in Lua
- Contributed bug fixes and features upstream to `fff` (10.8k stars), `snacks.nvim` (8.1k stars), and `blink.pairs` (296 stars)

Maintainer, qsyn - Quantum Circuit Synthesis Framework (github.com/DVLab-NTU/qsyn) Aug 2023 — Jun 2024
- Implemented C++20 quantum circuit optimization (198 stars) via ZX-calculus reduction and SAT-based oracle synthesis
- Co-authored the IEEE QCE '24 paper (`arXiv:2405.07197`) and maintained the core library with a 5-member research team
- Migrated the legacy build system to modern CMake and established automated Docker and GitHub Actions CI/CD

Team Lead, Light Dance Editor/Simulator (github.com/NTUEELightDance/LightDance-Editor)Nov 2022 — Mar 2023
- Led a 10-member team (240+ commits) to build a full-stack LED choreography editor, control UI, and Docker/ESLint CI
- Built a Three.js 3D stage simulator with custom GLSL shaders to optimize real-time LED strip rendering performance
- Collaborated with backend and embedded hardware teams to integrate live WebSocket controller and server protocols

Skills

- **Languages**: C/C++ (C++20), Rust, Go, Python, Lua, TypeScript/JavaScript, Kotlin, Swift, Nix, SQL, Bash/Fish
- **Systems & Embedded**: Linux Kernel (PCIe ASPM, PM, Kbuild), Android (AOSP, Soong, AIDL, SELinux), Modem BSP
- **Developer Tooling & Web**: Git/Jujutsu, Docker, CMake, Clangd, Neovim API, React, Three.js (GLSL), CI/CD